

HERNANDO, PANTINE B.
CHALLENGE 5 - CRYPTOGRAPHY

BSIT-3

Vigenere 



Medium

Cryptography

picoCTF 2022

AUTHOR: MUBARAK MIKAIL

Description

Can you decrypt this message?

Decrypt this [message](#) using this key "CYLAB".

Hints 

1

https://en.wikipedia.org/wiki/Vigen%C3%A8re_cipher

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This challenge is a medium-level cryptography task from **picoCTF 2022** that introduces the **Vigenère cipher**, a method of encrypting alphabetic text by using a series of interwoven Caesar ciphers.

Description

The challenge provides a ciphertext and a specific key required for decryption.

- **Ciphertext:** `rgnoDVD{O0NU_WQ3_G1G3O3T3_A1AH3S_f85729e7}`
- **Key:** `CYLAB`

The screenshot shows the dcode.fr website interface for the Vigenere Cipher. The left sidebar has a search bar, a list of tools, and social media links. The main area is titled 'VIGENERE CIPHER' and includes a 'VIGENERE DECODER' section with input fields for ciphertext, plaintext language, and alphabet. It also features a 'DECIPHERMENT METHOD' section with radio buttons for different decryption techniques. The right sidebar shows a 'VIGENERE ENCODER' section.

The hint provided points to a Wikipedia entry for the Vigenère cipher to help users understand the underlying poly-alphabetic substitution method.

Solution Strategy

To solve this, the ciphertext must be decrypted using the provided key. The Vigenère cipher uses the key to determine the shift for each letter in the message.

Steps to solve:

1. **Identify the Cipher and Key:** The description explicitly identifies the method as Vigenère and provides "CYLAB" as the key.
2. **Use a Decryption Tool:** A common approach is using an online tool like dcode.fr.
3. **Input Data:**
 - Enter the ciphertext: `rgnoDVD{00NU_WQ3_G1G3O3T3_A1AH3S_f85729e7}` into the decoder.

- Select the decryption method "Knowing the Key/Password" and enter CYLAB.
4. **Execute Decryption:** Clicking "Decrypt" processes the string.

Final Result

Once decrypted, the tool reveals the original plaintext flag.

Flag:

picoCTF{DONT_US3_V1G3N3R3_C1PH3R_d85729g7}